



UNIVERSITY COLLEGE OF SCIENCE AND TECHNOLOGY

INSTITUTE OF RADIO PHYSICS AND ELECTRONICS

REPORT ON MINI-PROJECT

TOPIC: DESIGN OF CLIPPERS AND CLAMPERS CIRCUIT USING DIODE

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We solemnly **declare** that the project is based on our own understanding. We assert the statements made and conclusions drawn are an outcome of our project work. We have further certify that:

- The work contained in the report is original and has been done by us.
- Whenever we have used materials (text, diagrams, etc) from other sources, we have given due credit to them in the text of the report and giving their details in bibliography

ABSTRACT

This project focuses on the design, simulation, and practical implementation of various clipper and clamper circuits using diodes, aimed at understanding their behaviour in signal conditioning applications.

Clipper circuits are used to remove or "clip" portions of input signals without distorting the remaining part, while clamper circuits shift the entire signal vertically without altering its shape.

The project includes implementation of both unbiased and biased clipper circuits, such as series and shunt configurations for both positive and negative clipping. Additionally, clamper circuits including positive, negative, and biased versions are designed. All circuits are simulated using circuit design tools Multisim

A combination of clipper and clamper circuits is also explored to demonstrate waveform shaping. This circuit is simulated using circuit design tools and then implemented on a custom-designed Printed Circuit Board (PCB) to demonstrate real-time behaviour. The project highlights the practical use of diodes, resistors, capacitors, and voltage sources in shaping and conditioning analog signals, which are essential in communication and electronic systems.

THEORY, WORKING PRINCIPLES WITH CIRCUIT DESIGNS

CLIPPING CIRCUITS: Clipping circuits are used for removing or cutting off portions of an input signal that go beyond a certain voltage level.

- Performed using diodes, resistors, and sometimes biasing voltages.
- Used to limit the amplitude of the signal.
- Output waveform is flattened above or below a threshold.
- It does not shift the signal — only limits certain parts.

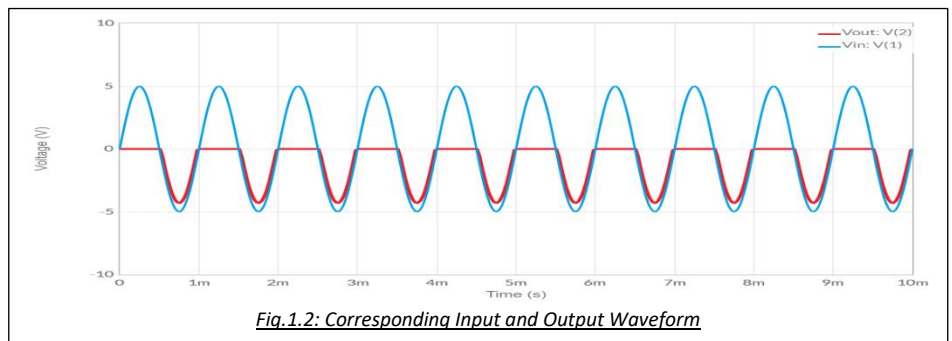
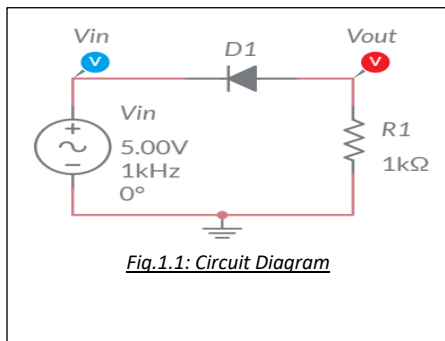
Applications:

- Signal protection (limiting voltage)
- Wave shaping in analog circuits
- Removing noise spikes

DIFFERENT TYPES OF CLIPPER CIRCUITS USING DIODES

(A) UNBIASED CLIPPER

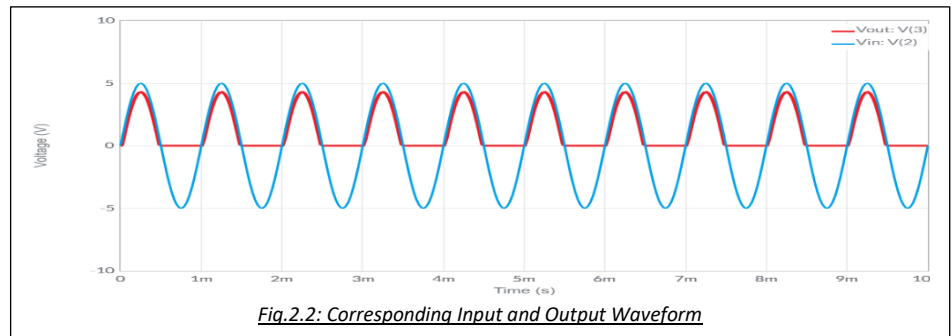
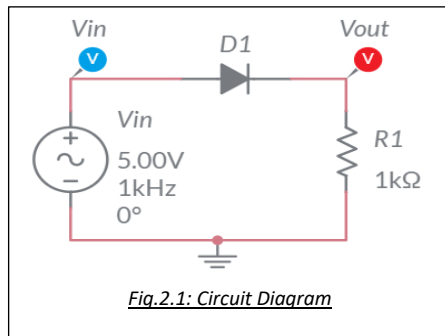
(1) Series Positive Clipper:



Diode in series with load. Clips positive half of the input waveform.

- During Positive Half: Diode becomes reverse-biased → blocks current → output is zero.
- During Negative Half: Diode is forward-biased → signal passes → output follows input.

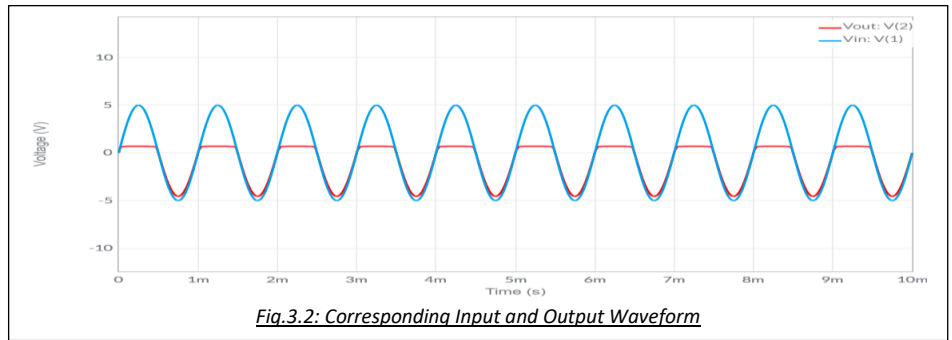
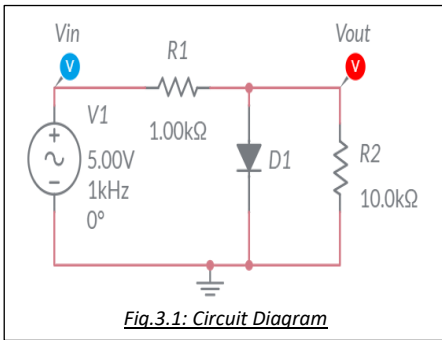
(2) Series Negative Clipper:



Diode in series with load(reversed). Clips negative half of the input waveform.

- During Positive Half: Diode becomes forward-biased → signal passes → output follows input.
- During Negative Half: Diode becomes reverse-biased → blocks current → output is zero.

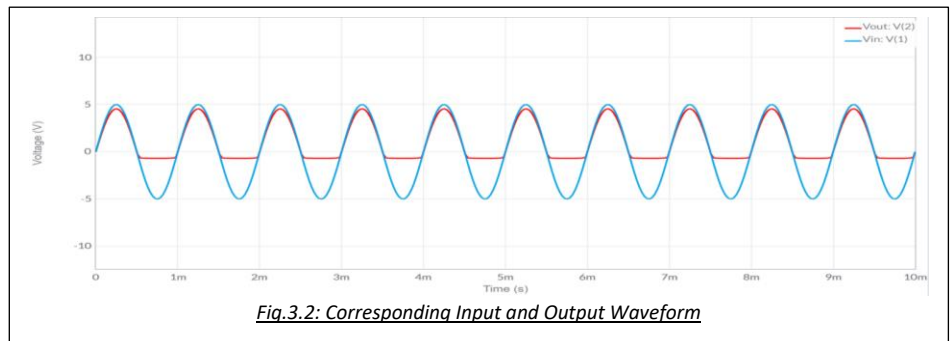
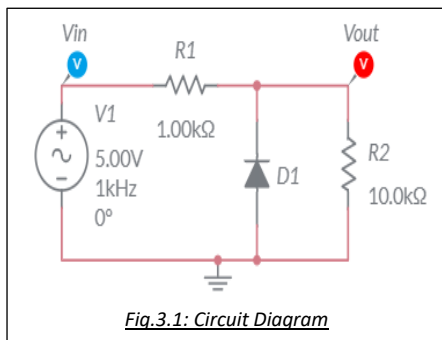
(3) Shunt Positive Clipper:



Diode connected in parallel (shunt) with load. Clips positive half by shorting it to ground.

- During Positive Half: Diode is forward-biased → conducts → signal is shorted to ground → output is zero.
- During Negative Half: Diode is reverse-biased → blocks → output follows input.

(4) Shunt Negative Clipper:

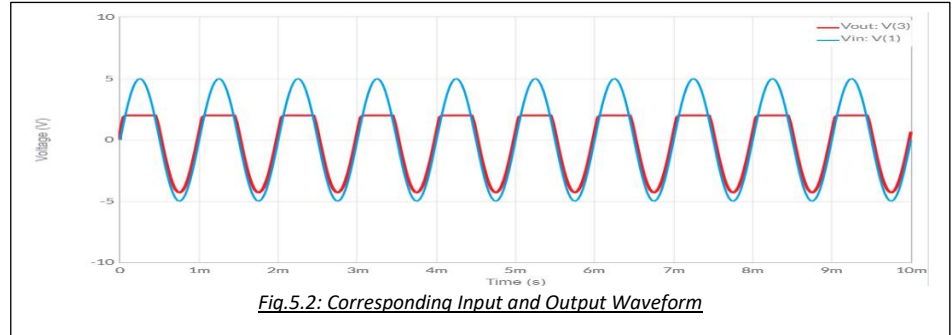
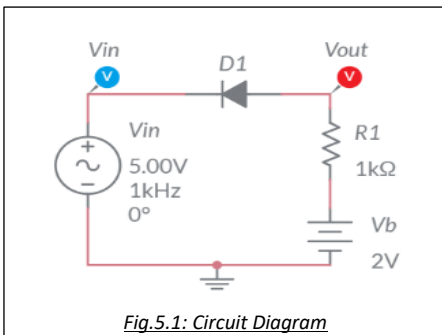


Diode connected in parallel (shunt) with load (reversed). Clips negative half of the waveform.

- During Positive Half: Diode is reverse-biased → blocks → output follows input.
- During Negative Half: Diode is forward-biased → conducts → signal is shorted to ground → output is zero.

(B) BIASED CLIPPER

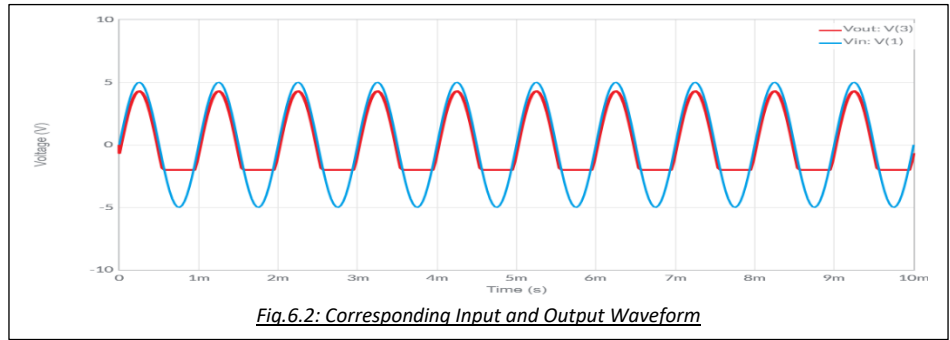
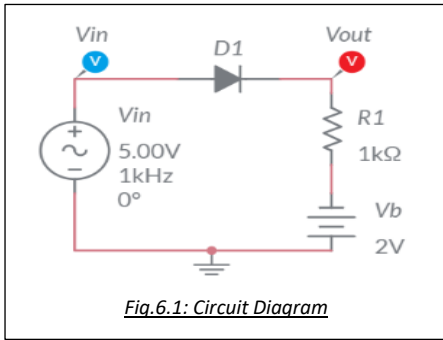
(1) Series Biased Positive Clipper:



Diode with positive bias in series with load. Clips signal above the bias level.

- During Positive Half: Diode is reverse-biased until input $> V_b + 0.7V = 2.7V$ → clips above this level.
- During Negative Half: Diode is forward-biased → signal passes → output follows input.

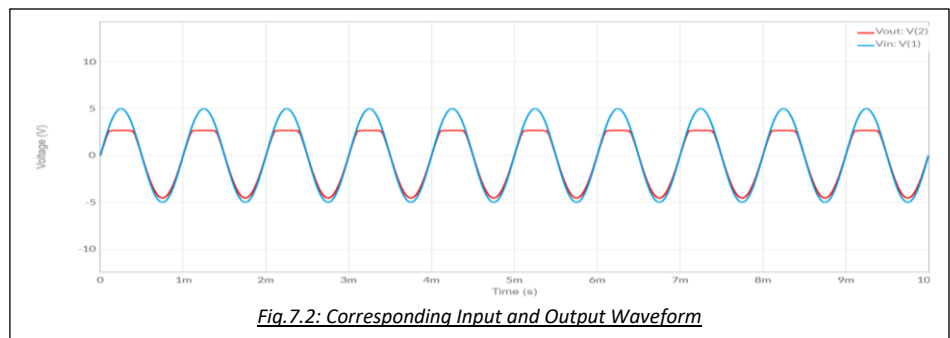
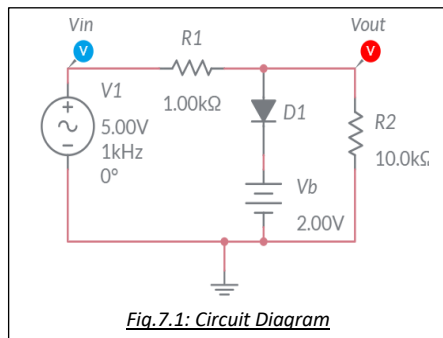
(2) Series Biased Negative Clipper:



Diode with negative bias in series (reversed). Clips signal below the negative bias level.

- During Positive Half: Diode is forward-biased → signal passes.
- During Negative Half: Diode is reverse-biased until input $< -(V_b + 0.7V) = -2.7V$ → clips below this level.

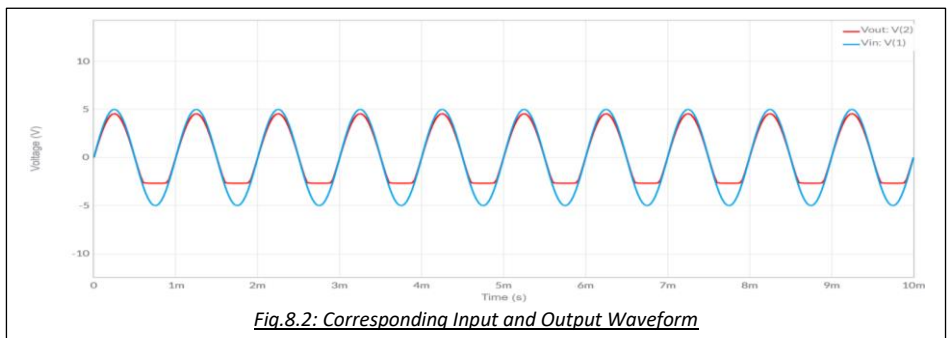
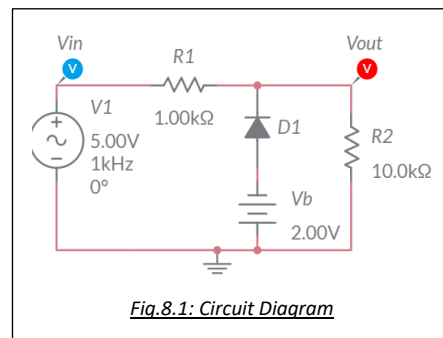
(3) Shunt Biased Positive Clipper:



Diode in shunt with positive bias. Clips signal above the bias voltage.

- During Positive Half: When input $> V_1 + 0.7V = 2.7V$ → diode becomes forward-biased → clips signal to bias level.
- During Negative Half: Diode is reverse-biased → signal passes → output follows input.

(4) Shunt Biased Negative Clipper:



Diode in shunt with negative bias (reversed). Clips signal below the bias voltage.

- During Positive Half: Diode is reverse-biased → output follows input.
- During Negative Half: When input $< -(V_1 + 0.7V) = -2.7V$ → diode conducts → signal is clipped to bias level.

CLAMPING CIRCUITS: Clamping circuits are used for shifting the entire signal up or down without changing its shape.

- Uses diodes, capacitors, and sometimes biasing voltages.
- The peak value remains the same, but the DC level changes.
- It adds a DC offset to the signal.
- Clamping moves the entire waveform above or below the zero level.

Applications:

- Restoring DC level in AC signals
- TV and oscilloscope signal processing
- Signal modulation and demodulation

DIFFERENT TYPES OF CLAMPER CIRCUITS USING DIODES

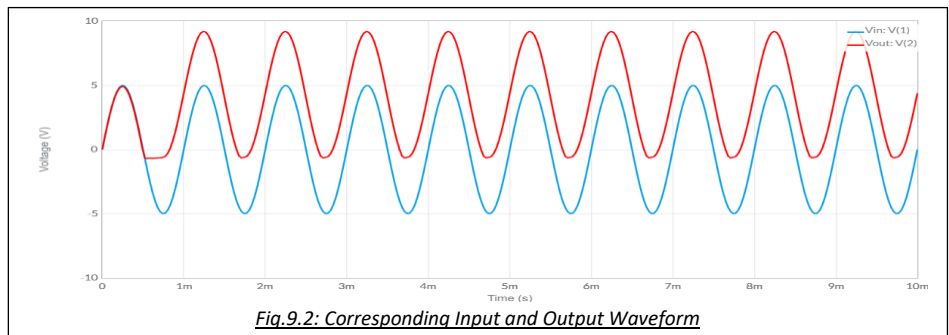
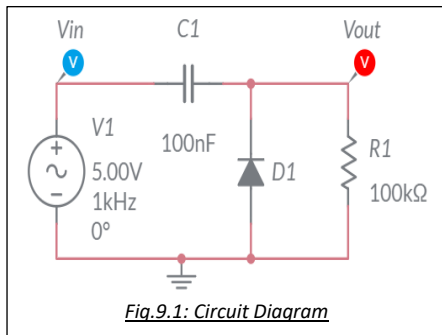
We have considered, Input frequency: 1 kHz; Load resistor: 100 k Ω

$$T = \frac{1}{1000} = 1\text{ms}$$

$$RC = 10T = 10 \times 1\text{ms} = 10\text{ms}$$

$$\text{hence, } C = \frac{10}{100000} = 0.1\mu\text{F} \quad \text{So, } C = \mathbf{0.1\mu\text{F}} \text{ is good for 1 kHz with 100 k}\Omega.$$

(A) POSITIVE CLAMPER CIRCUIT



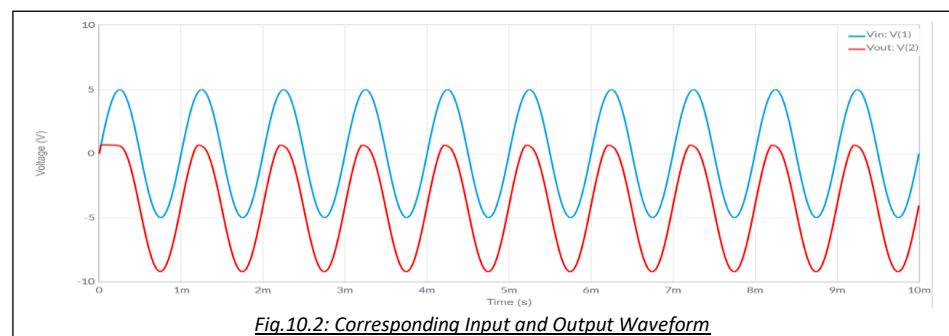
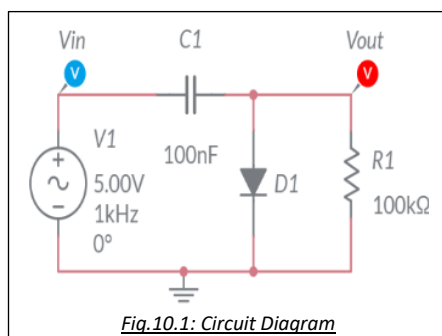
It consists of a diode, a resistor and a capacitor and that shifts the output signal to the positive portion of the input signal.

Initially, during the first positive half cycle input is given, the capacitor is not yet charged and the diode is reverse biased. During the negative half cycle, at the peak value, the capacitor gets charged to its peak value 5V. The diode is forward biased until the capacitor is fully charged and conducts current through it.

During the next positive half cycle, the capacitor is charged to positive 5V while the diode gets reverse biased and gets open circuited. The output of the circuit at this moment will be $V_{out} = V_{in} + 5V$.

Hence the signal is positively clamped as shown in the above fig.9.2.

(B) NEGATIVE CLAMPER CIRCUIT



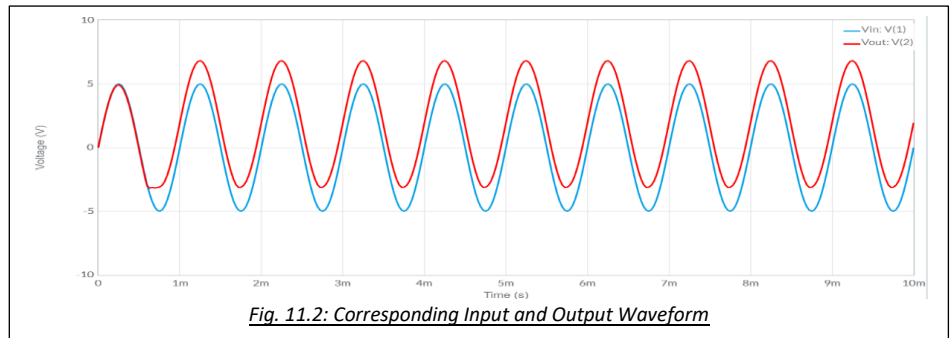
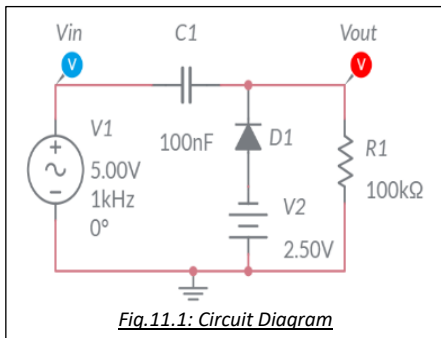
It consists of a diode, a resistor and a capacitor and that shifts the output signal to the negative portion of the input signal.

During the positive half cycle, the capacitor gets charged to its peak value -5V. The diode is forward biased and conducts during this period. During the negative half cycle, the diode gets reverse biased and gets open circuited.

The output of the circuit at this moment will be $V_{out} = -V_i - 5V$.

Hence the signal is negatively clamped as shown in the above fig.10.2

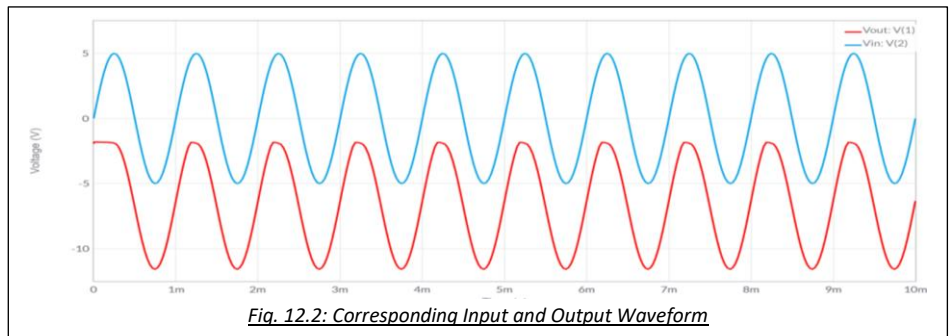
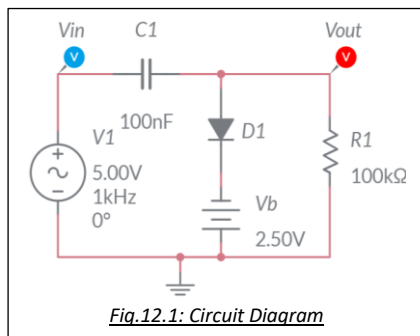
C) POSITIVE CLAMPER CIRCUIT WITH BIASING VOLTAGE



A Positive clamper circuit if biased with some negative reference voltage, that voltage will be added to the output to raise the clamped level.

During the positive half cycle, the voltage across the capacitor and the reference voltage together maintain the output voltage level. During the negative half-cycle, the diode conducts when the cathode voltage gets less than the anode voltage. These changes make the output voltage as shown in the above fig.11.2.

(D) NEGATIVE CLAMPER CIRCUIT WITH BIASING VOLTAGE



A Negative clamper circuit if biased with some negative reference voltage, that voltage will be added to the output to raise the clamped level.

The cathode of the diode is connected with a negative reference voltage, which is less than that of zero and the anode voltage. Hence the diode starts conducting during positive half cycle, before the zero-voltage level. During the negative half cycle, the voltage across the capacitor appears at the output. Thus, the waveform is clamped towards the negative portion as shown in the above fig.12.2.

NOTE: Choosing the correct resistance value (R) in clipper circuits — whether series or shunt, positive or negative, biased or unbiased — depends on the circuit's desired behavior, diode specifications, and the input signal properties.

FOR SERIES CLIPPER CIRCUIT:

Choose a moderate to high value

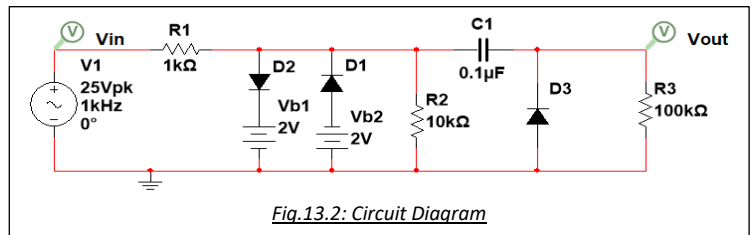
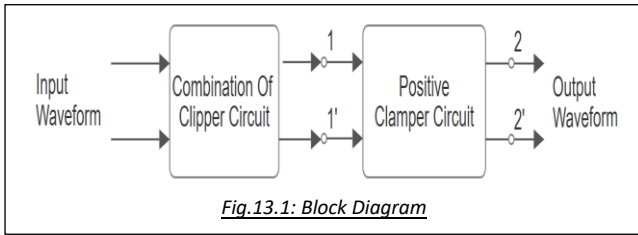
- To limit the current when the diode conducts.
- To ensure voltage drop appears mainly across the diode (during conduction).
- Resistance should not be too high, else the signal will be severely attenuated.

FOR SHUNT CLIPPER CIRCUIT:

Choose a relatively low value

- To provide a path for current when diode conducts.
- To develop enough voltage drop across it for proper clipping.
- If Resistance is too large, the diode might not get enough forward voltage to turn ON.

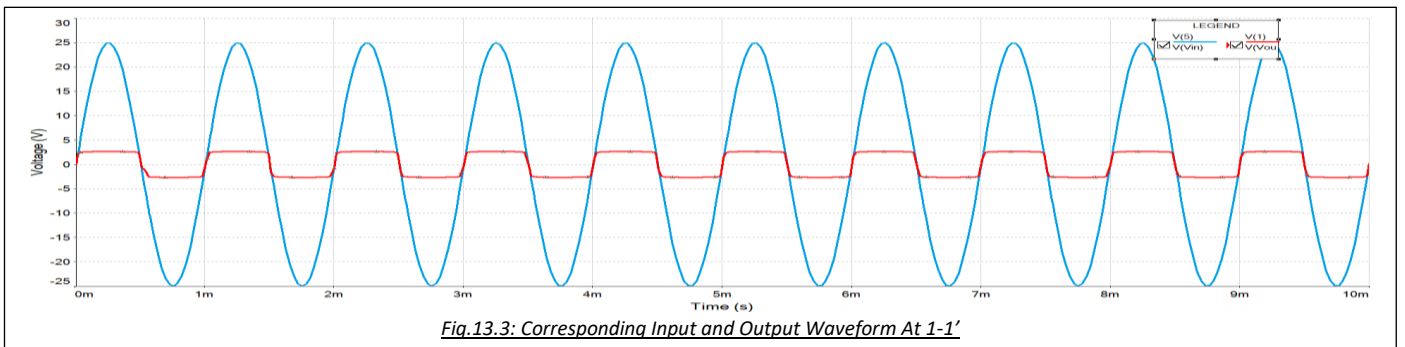
DESIGN AND ANALYSIS OF CLIPPER-CLAMPER CIRCUITS FOR SIGNAL CONDITIONING OF SINUSOIDAL SIGNALS



We have taken a sinusoidal input band setting its peak to peak voltage at 25V, frequency of 1KHz and 0 phase delay. The circuit block diagram is given in fig.13.1 with its corresponding circuit diagram in fig13.2.

Now let us look into the 1st block i.e. the combination of clipper circuit. When a portion of both positive and negative of each half cycle of the input voltage is to be clipped (or removed), combination clipper is employed which combines two diodes and two bias voltages in opposite directions. Clips both the positive and negative ends of the waveform at different levels. It is used to limit signals between two voltage thresholds.

For positive input voltage signal when input voltage exceeds battery voltage $V_{b1}(=+2V)$ diode D_2 conducts heavily while diode D_1 is reversed biased and so voltage $V_1(=+2V)$ appears across the output resistance $R_2(=10k\Omega)$. This output voltage $V_{b1}(=+2V)$ stays as long as the input signal voltage exceeds $V_{b1}(=+2V)$. On the other hand, for the negative input voltage signal, the diode D_2 remains reverse biased and diode D_1 conducts heavily only when input voltage exceeds battery voltage $V_{b2}(=-2V)$ in magnitude. Thus, during the negative half cycle the output stays at $V_{b2}(=-2V)$ so long as the input signal voltage is greater than $V_{b2}(=-2V)$.

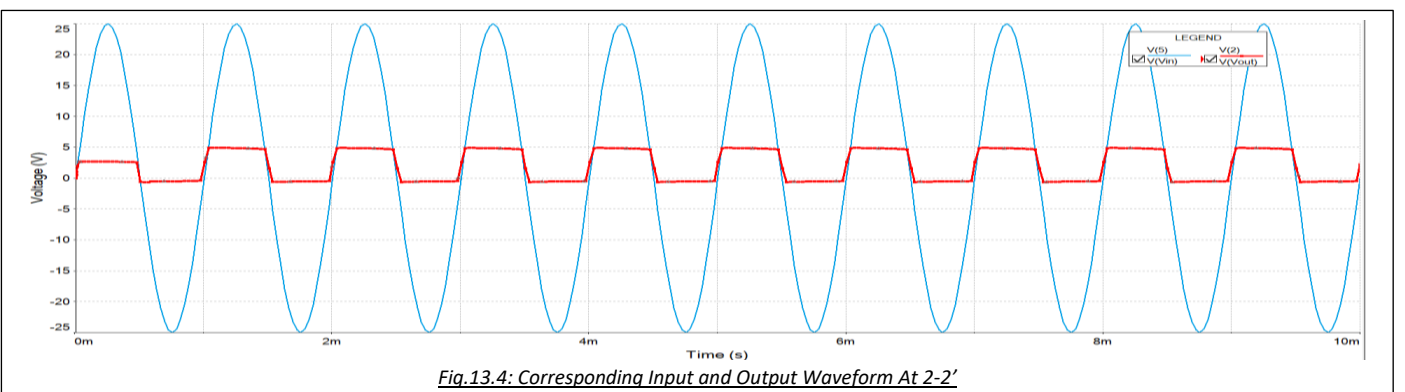


The output of the combination of clipper circuit is shown in fig.13.3. This output is feed as input to our next block i.e. positive clamper circuit.

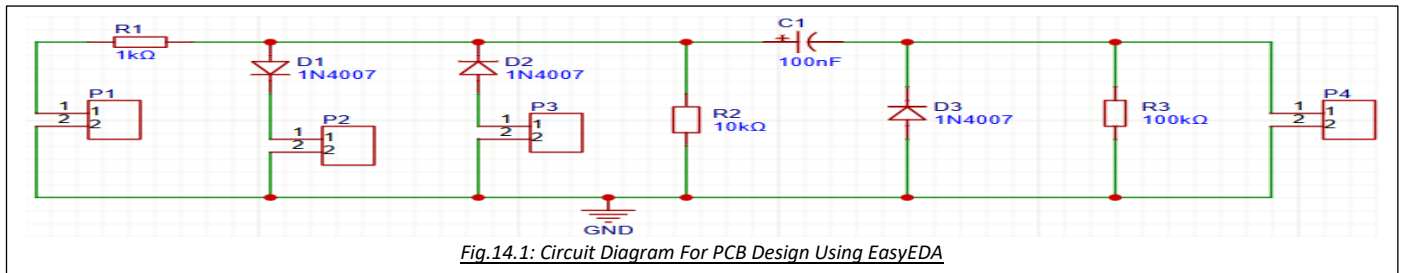
Initially, during the first positive half cycle input is given, the capacitor C_1 is not yet charged and the diode D_3 is reverse biased. During the negative half cycle, at the peak value, the capacitor gets charged to its peak value 2V. The diode is forward biased until the capacitor is fully charged and conducts current through it.

During the next positive half cycle, the capacitor is charged to positive 2V while the diode gets reverse biased and gets open circuited. The output of the circuit at this moment will be $V_{out}=V_{in}+2V$.

Hence the signal is positively clamped as shown in the above fig.13.4.



COMPONENTS REQUIRED:



- SCREW TERMINAL BLOCKS (P1,P2,P3&P4)
- FUNCTION GENERATOR
- C.R.O(Cathode Ray Oscilloscope)
- SINGLE STRANDED WIRE
- PCB (Printed Circuit Board)

FOR CLIPPING CIRCUIT

- AC VOLTAGE SOURCE
- RESISTANCE – 1KΩ & 10KΩ
- TWO DIODES (D1 & D2)
- TWO 2V DC SOURCE (Vb1 & Vb2)

FOR CLAMPER CIRCUIT

- CAPACITOR – 0.1μF
- RESISTANCE – 100KΩ
- DIODE(D3)

PRINTED CIRCUIT BOARD (PCB) DESIGN

A PCB mechanically supports and electrically connects electronic components using conductive tracks, pads and other features etched from sheets laminated on to a nonconductive substrate.

A PCB consists of a conducting layer that is made up of copper lamina. The insulating layer dielectric is laminated together with epoxy resin. Boards may be single layered, double-layer or multi-layered. In this project we have used single layered PCB.

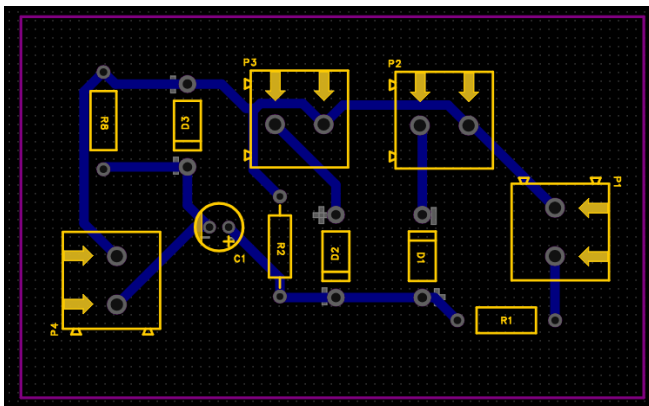


Fig.15.1: PCB Design Layout Using EasyEDA

PCB LAYOUT DESIGNING

PCB layout consists of the holes, tracks, pads, via, components, etc positions in a given PCB design. This can be done directly by hand on the PCB using a marker, if the design is very simple, otherwise we need to take help of a software. We have used EasyEDA software for PCB layout designing. Today's PCB which consists of multiple layers are very complex and can be designed using software only.

PCB ETCHING PROCESS

All PCBs are made by bonding a layer of copper onto the entire substrate. The etching process is used to remove the unwanted copper, leaving only the desired copper tracks. This is done after applying a temporary mask over the copper layer. Although there are various etching methods available, we used the most common one. First, we took a printout of the mirror image of the circuit design on photo paper. Then, we transferred the layout onto the PCB board using a heated iron. Finally, we etched the board using a Ferric Chloride solution.

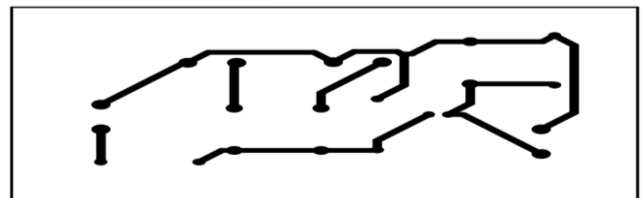
$$2\text{FeCl}_3 + \text{Cu} \rightarrow 2\text{FeCl}_2 + \text{CuCl}_2$$


Fig.15.2: Mirrored Image Of Circuit Design Layout For Temporary Mask



Fig.15.3: Etching By Ferric Chloride Solution

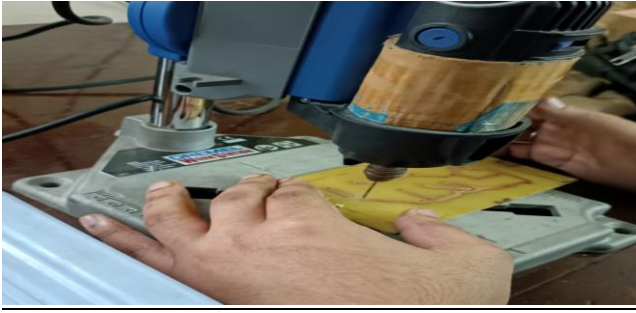


Fig.15.4: PCB Drilling

PCB DRILLING

The components that have to be attached to the PCB is done by hole drilling. For a multi-layered PCB via drilling may be needed to connect tracks which are present on different layers. A Dremel drill press is used to punch the holes with micro drill bits. For high volume production automated drilling machines are used

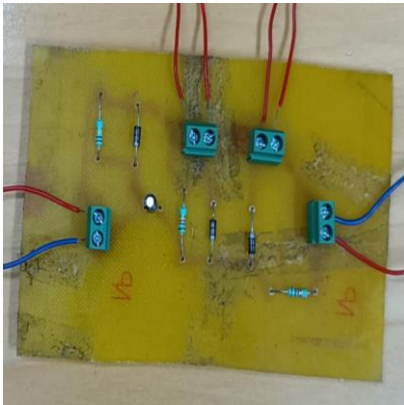


Fig.15.5: Top View (Components On PCB)

PCB ASSEMBLING

PCB assembling includes the assembling of the electronic components on the respective holes in the PCB. The components leads are then soldered to the PCB using soldering equipments. The common aspect of PCB assembling is that the components leads are electrically connected and mechanically fixed to the board using a molten metal solder

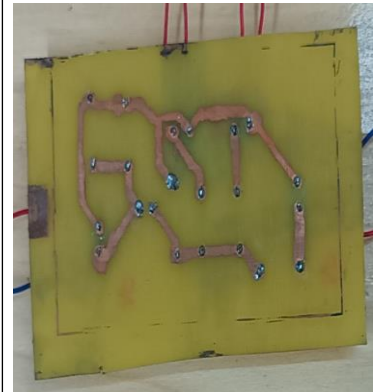


Fig.15.6: Bottom View (Connecting Tracks)

APPLICATIONS

(1) Overvoltage Protection for Sensitive Devices

- Maintaining the signal levels within desired dynamic range
- By clipping the signal at ± 2 V, we protect downstream circuits (like op-amps or microcontroller inputs) from higher voltage spikes

(2) Analog-to-Digital Converter (ADC): This circuit ensures:

- The signal does not exceed upper/lower voltage limits (due to clipping)
- The signal is shifted into the ADC's input range (due to clamping)

RESULT AND DISCUSSION

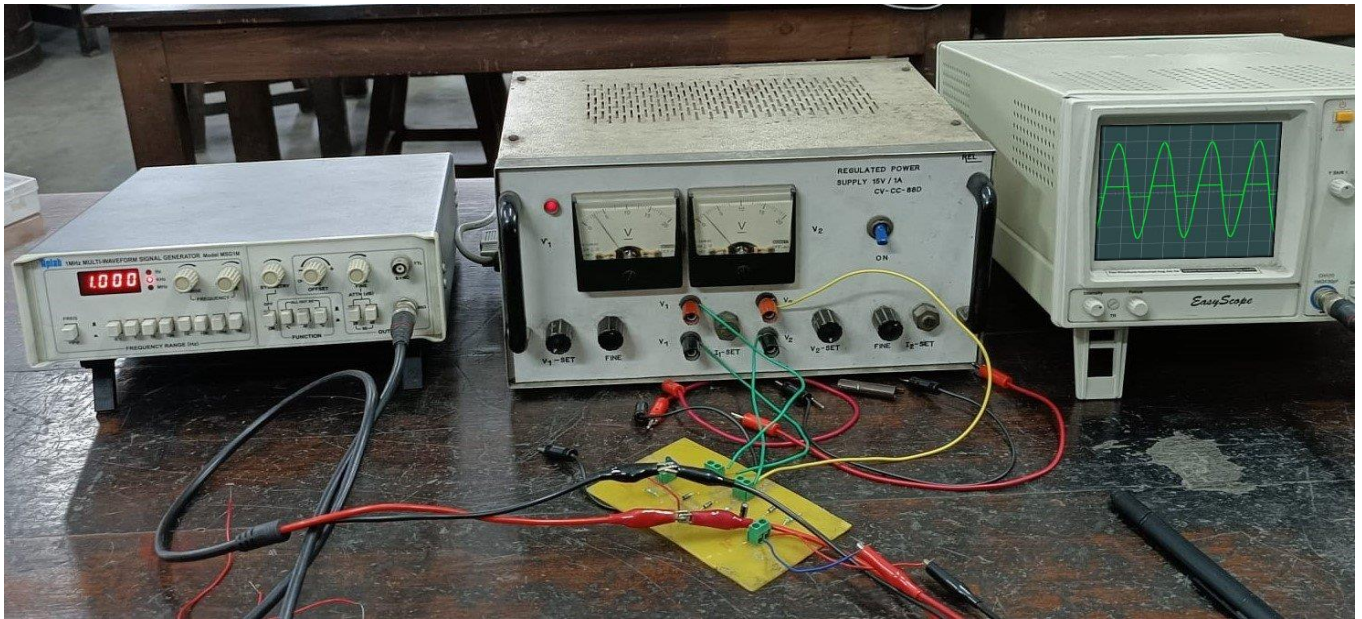


Fig.16.1: Working Of The PCB Design In Real Time

CONCLUSION: The clipper-clamper circuit was successfully designed, implemented, and tested on a PCB. It effectively clipped the input sinusoidal waveform and shifted it upward using a positive clamper. The real-time output matched the expected results, confirming proper operation.

PRECAUTIONS AND DISCUSSIONS:

- (1) 90° trace angles are avoided in PCB designs as it can cause impedance discontinuities, leading to signal reflections. Moreover, in chemical etching during PCB fabrication, sharp inside corners like those in 90° angles can trap etchant leading to broken traces
- (2) If the circuit transferred from the photopaper to the copper board is incomplete or has broken tracks, you can use nail polish (or a permanent marker) to manually cover (mask) the missing or broken areas before etching.
- (3) During the experiment, it is important to ensure that the diodes are connected in the correct orientation and the bias voltages are accurately applied at ± 2 V. The capacitors polarity must be maintained properly. All circuit connections should be tight and secure to avoid fluctuations or incorrect outputs. A regulated power supply is recommended to maintain stable voltage levels. The function generator should be set to provide a 25 V peak, 1 kHz sinusoidal input. Finally, sufficient time should be allowed for the capacitor in the clamper circuit to fully charge before observing the output waveform

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 - a. https://www.youtube.com/watch?v=S76CnEJMI5E&list=PLwjK_ikyK4LLBVM18VZ7JKW-q88FAtnr8_&index=48
 - b. https://www.youtube.com/watch?v=7O3Hbkk624&list=PLwjK_ikyK4LLBVM18VZ7JKW-q88FAtnr8_&index=50

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THE END